

TravelShare Memory Map Feature – Technical Development Plan

Overview

A programmer would typically build the **Memory Map** feature using four core components:

1. Location Capture
2. Map Display
3. Memory Storage
4. Trip Visualization

The goal is to transform user-generated travel memories into an interactive, map-based journey that allows travelers to relive their experiences geographically.

Architecture Overview

User Takes Photo

↓

GPS Coordinates Captured

↓

Photo Uploaded

↓

Database Stores:

- Image URL
- Latitude
- Longitude
- Date
- Contributor ID

↓

Map API Displays Pins

↓

User Taps Pin



Photo Gallery Opens

Recommended Technology Stack

Mobile App

Recommended frameworks:

- Flutter
- React Native

Both support:

- Google Maps
- Apple Maps
- Mapbox

Backend

Recommended options:

- Firebase
- Node.js + PostgreSQL

Storage

- Firebase Storage
- AWS S3

Maps

Recommended: Mapbox

Benefits:

- Beautiful tourism-focused map styles
- Custom map styling
- Offline map capabilities
- Lower cost at scale than Google Maps
- Excellent route visualization features

Database Structure

Trips Table

```
{
  "trip_id": "12345",
  "user_id": "abc123",
  "trip_name": "Barbados Vacation",
  "start_date": "2026-06-01",
  "end_date": "2026-06-07"
}
```

Memories Table

```
{
  "memory_id": "m001",
  "trip_id": "12345",
  "image_url": "...",
  "latitude": 13.0968,
  "longitude": -59.6145,
  "location_name": "Carlisle Bay",
  "created_at": "2026-06-02",
  "approved": true
}
```

Step 1: Capture Location

When a tourist uploads a memory, the application captures the device's GPS coordinates.

Example (Flutter):

```
Position position =
  await Geolocator.getCurrentPosition();

double lat = position.latitude;
double lng = position.longitude;
```

Store:

```
{  
  "latitude": 13.0968,  
  "longitude": -59.6145  
}
```

Step 2: Reverse Geocoding

Convert GPS coordinates into a readable destination name.

Input:

```
13.0968, -59.6145
```

Output:

```
Carlisle Bay, Barbados
```

Services:

- Google Geocoding API
- Mapbox Geocoding API

Store both coordinates and place names.

```
{  
  "latitude": 13.0968,  
  "longitude": -59.6145,  
  "location_name": "Carlisle Bay"  
}
```

Step 3: Display Pins on the Map

The map loads all approved memories and displays them as interactive markers.

Example:

```
memories.map(memory => {  
    addMarker(  
        memory.latitude,  
        memory.longitude  
    );  
});
```

Results:

-  Carlisle Bay
 -  Oistins
 -  Bathsheba
 -  Animal Flower Cave
-

Step 4: Open Memory Galleries

When a user taps a map marker:

```
onMarkerClick(memory)
```

Display:

Carlisle Bay

- 15 Photos
- 3 Videos
- June 2, 2026

Action:

Open Gallery

The user can then view all photos and videos associated with that location.

Step 5: Create Journey Routes

To visualize the travel experience, memories are organized chronologically.

Example:

9:00 AM – Carlisle Bay

11:30 AM – Harrison's Cave

2:00 PM – Bathsheba

Mapbox can draw a route connecting these locations.

```
LineString([  
  [lng1, lat1],  
  [lng2, lat2],  
  [lng3, lat3]  
])
```

Users see a travel route similar to Google Timeline.

Step 6: Build a Memory Heatmap

Count memory activity by location.

Example:

Carlisle Bay = 42 Photos

Oistins = 31 Photos

Bathsheba = 12 Photos

Using Mapbox Heat Layers:

```
heatmap-weight
```

This visually highlights the places where the user created the most memories.

Feature Name:

Your Most Memorable Places

Step 7: Trip Replay Feature

Users can select:

▶ Replay My Trip

The application animates the journey:

Day 1 → Pins Appear

Day 2 → Additional Pins Appear

Day 3 → Journey Continues

The map zooms across Barbados while displaying memories captured at each stop.

Think of it as:

Spotify Wrapped meets Google Earth.

Step 8: Island Passport Gamification

Create tourism zones across Barbados.

Example:

```
{  
  "zone": "Bridgetown"  
}
```

When users upload content in a new zone:

Achievement Unlocked

✓ Bridgetown Explorer

Milestones:

- 10 Locations Visited → Island Adventurer
- 20 Locations Visited → Barbados Master Explorer

This encourages visitors to explore more of the island.

Privacy Controls

Because privacy is a core TravelShare value, location sharing must be optional.

Exact Location

```
{  
  "latitude": 13.09681,  
  "longitude": -59.61452  
}
```

Approximate Location

```
{  
  "latitude": 13.09,  
  "longitude": -59.61  
}
```

Hidden Location

```
{  
  "region": "South Coast"  
}
```

Only a general area is displayed.

Future AI Enhancement

The platform can automatically generate travel journals from collected memories and map data.

Example:

"On June 2nd you explored Carlisle Bay, snorkeled with turtles, and captured 18 memories from fellow travelers."

This turns photo collections into personalized travel stories.

Recommended Version 1 Launch Features

To deliver maximum impact quickly, the first release should include:

1. Mapbox Map Integration
2. Memory Location Pins
3. Photo and Video Galleries
4. Journey Route Visualization
5. Trip Replay Animation

These five features would immediately distinguish TravelShare from Google Photos, Instagram, and Apple Photos by creating a travel-focused memory experience centered around location, storytelling, and exploration.